

**EDGE INTELLIGENCE
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CASE STUDY II**

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EDGE VIDEO ANALYTICS

Abstract:

Edge Video Analytics (EVA) represents a transformative convergence of edge computing, deep learning, and real-time computer vision. By relocating computation from centralized cloud data centers to geographically distributed edge nodes, EVA enables video processing with dramatically reduced latency, lower bandwidth consumption, and stronger privacy guarantees. This write-up provides a comprehensive technical review of the EVA landscape, drawing on twelve peer-reviewed research papers published between 2020 and 2025. It examines the fundamental motivations for EVA, the major system architectures including edge-only, hierarchical edge-cloud, and collaborative multi-edge topologies, and the suite of enabling techniques that make low-power real-time inference feasible: model compression (pruning and quantization), knowledge distillation, neural architecture search, dynamic task offloading, and adaptive video encoding. The review covers the principal application domains, discusses open research challenges around accuracy-efficiency trade-offs, hardware heterogeneity, privacy, and security, and concludes with an outlook on emerging directions such as federated learning at the edge and transformer-based video understanding.

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1. Introduction

Video is the dominant medium of modern digital data. Cisco's Visual Networking Index estimated that video traffic accounted for over 80 % of all internet traffic even before the proliferation of IoT cameras. Today, governments and enterprises deploy millions of surveillance cameras for law enforcement, traffic management, industrial quality control, smart-retail analysis, and emergency response. Analyzing these streams in real time requires inference of computationally intensive deep neural networks (DNNs) on every captured frame.

Traditionally, video analytics has been a cloud-centric endeavor: raw streams are transmitted over wide-area networks to centralized clusters, processed by large server-grade GPUs, and results are returned to end-users. This model suffers from three fundamental limitations. First, the round-trip latency over the WAN can exceed hundreds of milliseconds, violating the sub-100 ms requirements of real-time control applications such as autonomous vehicles or industrial robotics. Second, transmitting high-definition video from thousands of cameras generates enormous backhaul bandwidth costs, often constituting the dominant operational expense of a smart-city deployment. Third, raw video contains sensitive biometric information; transmitting it to a cloud service introduces significant privacy and regulatory compliance concerns.

Edge computing addresses these limitations by placing compute, storage, and networking resources at the periphery of the network, in close physical proximity to data sources. The intersection of edge computing and video analytics has given rise to the active research domain of Edge Video Analytics (EVA). Xu et al. (2023) define EVA as "the deployment of video analytics systems that leverage edge computing resources to process video data locally or semi-locally, enabling real-time or near-real-time inference with reduced latency and bandwidth consumption" [1]. The field has grown explosively, with IEEE Communications Surveys & Tutorials publishing a comprehensive survey cataloging over 200 recent contributions [1].

This write-up tells the current state of EVA across eight dimensions: (i) system architectures, (ii) model compression, (iii) knowledge distillation, (iv) adaptive video encoding, (v) task offloading, (vi) hardware acceleration, (vii) application domains, and (viii) open challenges. Throughout, it grounds claims in recent primary literature published between 2020 and 2025.

2. Background and Motivation

2.1 The Limits of Cloud-Centric Video Analytics

A canonical cloud-centric video pipeline operates as follows. An IP camera encodes video at 1080p at 30 frames per second (fps), transmits the H.264 stream over a LTE or fibre uplink, and a cloud inference server runs object detection DNN (e.g., Faster R-CNN or YOLO) on each frame. This pipeline introduces three bottlenecks:

- **Latency:** WAN round-trip times of 50–300 ms make real-time feedback impossible for safety-critical applications.
- **Bandwidth:** A single 1080p 30 fps stream consumes roughly 8–20 Mbps; a city-scale deployment with 10,000 cameras exceeds 100 Gbps of uplink demand.
- **Privacy:** Unencrypted raw video frames contain faces, license plates, and behavioral patterns; transmitting them off-site violates GDPR and similar data protection laws.

The 2023 survey by Xu et al. on EVA summarizes these pain points rigorously: "cloud-based solutions for deep learning are dependent on reliable network access, and it is challenging to offload the full or partial compute tasks to the proximate edge given latency and privacy constraints" [1].

2.2 Edge Computing Fundamentals

Edge computing moves workloads closer to where data is generated. The compute tier may be an on-camera processor (embedded GPU, NPU), a roadside micro-server (NVIDIA Jetson AGX, Intel Neural Compute Stick), or a multi-access edge compute (MEC) node co-located with a 5G base station. These platforms provide several orders of magnitude less compute than cloud clusters (typically 1–10 TFLOPS vs. thousands in the cloud) but offer sub-millisecond local processing latency.

Research by Naveen and Kounte (2024) in the Journal of Engineering Science and Technology Review underscores the hardware context: "applications like Smart Traffic

Management and autonomous vehicles involve deploying edge devices with cameras and sensors at intersections and roadsides to monitor and analyze real-time traffic conditions" and demand on-device inference solutions that are both accurate and power-efficient [3].

2.3 The Role of Deep Learning

The accuracy revolution in computer vision has been driven almost entirely by deep convolutional neural networks. Models such as ResNet, EfficientDet, YOLO, and Vision Transformers now match or exceed human-level accuracy on standard object detection and action recognition benchmarks. However, these models carry large computational footprints: a standard ResNet-50 requires approximately 4 GFLOPs per inference and 25 million parameters. Running such models at 30 fps on a camera with a 5W power budget is infeasible without compression or offloading. This tension between accuracy and efficiency is the central engineering challenge of EVA and motivates the entire sub-field of model compression reviewed in Section 4.

3. System Architectures for Edge Video Analytics

EVA systems have converged on three broad architectural patterns, each trading off deployment simplicity, scalability, latency, and accuracy differently. The 2023 arXiv survey on edge-based video analytics [2] provides the most systematic taxonomy of these patterns.

3.1 Edge-Only Architecture

In an edge-only system, all video capture, encoding, inference, and actuation logic reside on the edge device. There is no dependency on a cloud back-end. A representative deployment is an NVIDIA Jetson Nano or Raspberry Pi 4 camera unit running a quantized YOLOv8-nano to detect pedestrians. The advantages are minimal latency (typically < 20 ms end-to-end), complete data sovereignty, and graceful operation during network outages. The disadvantage is that edge processors impose

strict constraints on model complexity, inference throughput, and the ability to update or retrain models.

The survey by Ke et al. (2023) [2] notes that in edge-only deployments, "deep learning algorithms are often computationally intensive, while the front-end equipment usually lacks the computing power for executing large-scale deep learning tasks." This observation motivates the extensive literature on model compression surveyed in Section 4.

3.2 Hierarchical Edge-Cloud Architecture

The hierarchical architecture, illustrated in Figure 1 below, is the most commonly adopted pattern in practice. It comprises three tiers: (i) end devices (cameras and IoT sensors), (ii) edge servers (local micro-data centers, MEC nodes), and (iii) cloud servers (hyperscale clusters). Tasks are partitioned across these tiers based on latency requirements, computational intensity, and data sensitivity.

A lightweight detection model on the camera performs quick first-pass filtering (e.g., motion detection, low-confidence object proposals). Confirmed regions of interest are forwarded to the edge server running a heavier model. Complex analytics requiring large context windows (e.g., long-term behavior modeling, model retraining) are offloaded to the cloud. Xu et al. (2023) confirm that the hierarchical architecture "has been verified as the most efficient and scalable one for edge computing deployments" [1].

The REACT system (Ghosh et al., 2023), presented at ACM IoTDI 2023, is a canonical example [7]. REACT deploys a lightweight SSD-MobileNetV2 on the edge and a heavy Faster R-CNN or CenterNet on the cloud. Whenever the edge model has low confidence, the frame is asynchronously forwarded to the cloud for a second-pass inference. The cloud's result then retroactively updates the edge analytics pipeline. On the VisDrone benchmark, REACT outperforms edge-only baselines by up to 50% mAP.

3.3 Collaborative Multi-Edge Architecture

In a multi-camera deployment such as a smart city intersection cluster, multiple edge nodes can pool resources to collaboratively execute analytics tasks that no single node could handle alone. Collaborative multi-edge systems exploit spatial redundancy across camera views and temporal redundancy across frames to reduce total computation. CrossRoI (Guo et al., 2021) demonstrated cross-camera region-of-interest optimization

that reduces redundant computation when multiple cameras cover overlapping fields of view [1].

The BiSwift system (2023) [6] exemplifies multi-stream edge orchestration. It uses deep reinforcement learning to dynamically allocate bandwidth and select inference pipelines for multiple simultaneous video streams sharing an edge node. By co-optimizing bandwidth allocation and pipeline selection, BiSwift achieves superior accuracy-latency trade-offs compared to per-stream greedy baselines.

Dong et al. (2022) proposed a multi-agent deep reinforcement learning approach for collaborative video analytics on distributed edges [1]. Each edge agent learns to route analytics tasks to neighbors when local resources are congested, achieving load balancing without a centralized controller.

Architecture	Latency	Bandwidth Usage	Privacy	Scalability
Edge-Only	< 20 ms	None (local)	Maximum	Limited
Hierarchical Edge-Cloud	20–200 ms	Low (features only)	High	High
Collaborative Multi-Edge	< 50 ms	Medium (peer links)	High	Very High

Table 1: Comparison of EVA System Architectures

4. Enabling Techniques

4.1 Model Compression: Pruning and Quantization

Model compression is the primary technique for bridging the gap between accuracy-hungry DNN models and resource-constrained edge hardware. Two complementary families of methods dominate the literature: structured/unstructured pruning and fixed-point quantization.

4.1.1 Pruning

Pruning removes redundant parameters from trained networks to reduce model size and inference latency. Unstructured pruning zeros out individual weights based on magnitude thresholds; it can achieve high compression ratios (often > 90%) but does not translate directly to wall-clock speedup without sparse-compute hardware support. Structured pruning, by contrast, removes entire filters, channels, or layers, yielding dense compressed models that run faster on commodity hardware.

The survey by Liang et al. (2023) in MDPI Computers [5] synthesizes state-of-the-art compression methods: "model compression has become a focus of research... techniques play an important role in deploying models on edge devices... this paper summarized the state-of-the-art techniques, including model pruning, parameter quantization, low-rank decomposition, and knowledge distillation." In the paper, structured pruning is shown to reduce model sizes by up to 75% with less than 2% accuracy degradation on standard benchmarks.

Fang et al.'s DepGraph (2023) demonstrated architecture-agnostic structured pruning, achieving 50% inference speedup on CNNs while maintaining accuracy within a 1% margin, making it directly applicable to real-time video pipelines [5].

4.1.2 Quantization

Quantization replaces 32-bit floating-point parameters with lower-precision integers (INT8, INT4, or even binary). INT8 quantization roughly halves memory footprint and doubles throughput on hardware with native integer execution units (e.g., NVIDIA TensorRT, Google Edge TPU). Post-training quantization (PTQ) applies quantization after training with minimal accuracy loss; quantization-aware training (QAT) incorporates quantization error into the training objective for higher final accuracy.

Research by Naveen and Kounte (2024) [3] demonstrated that combining structured pruning with 8-bit quantization on ResNet-34 and VGG-16 achieves accuracy within 1% of the uncompressed baseline while dramatically reducing model size. Their framework achieved 78.01% accuracy for ResNet-34 after compression, compared to a 79.01% baseline—a marginal loss of < 1%. Deployed on a resource-constrained IoT edge cluster, the compressed model achieved an inference time of 20 ms with 92.5% accuracy [3].

When pruning and quantization are combined, compression rates are multiplicative. Experiments in the comprehensive model compression survey by Frontiers in Robotics and AI (2025) [5] show that OTov3 pruning followed by 8-bit dynamic quantization achieves an 89.7% reduction in model size and 95% reduction in parameters/MACs,

while the model accuracy actually increases by 3.8%, likely due to regularization effects of the compression.

4.2 Knowledge Distillation

Knowledge distillation (KD), introduced by Hinton et al. (2015) and extensively developed since, trains a compact “student” model to mimic the soft output probability distributions (“dark knowledge”) of a large “teacher” model, rather than training on hard one-hot labels alone. This allows students to achieve higher accuracy than models trained from scratch at the same parameter count.

In the EVA context, a heavy cloud-resident teacher model (e.g., a 100M-parameter Vision Transformer) produces predictions on labeled video frames, and a lightweight edge-resident student model (e.g., a 3M-parameter MobileNet) learns from both the ground-truth labels and the teacher’s soft outputs. Ruiz-Barroso et al. (2025) [8] demonstrated exactly this pattern in real-time unsupervised video object detection on embedded edge hardware, combining KD with FP16 precision and concurrent multi-unit task deployment to achieve a substantial reduction in computational complexity without penalizing the original mAP of 59.44.

Li et al.’s survey “When Object Detection Meets Knowledge Distillation” (IEEE TPAMI, 2023) [4] provides a systematic taxonomy of KD for detection, covering feature-level distillation (matching intermediate feature maps between teacher and student), relation-based distillation (preserving inter-sample relationships), and response-based distillation (matching final predictions). Feature-level methods such as DeCoupled Knowledge Distillation (DKD) and CrossKD consistently outperform response-only distillation for dense detection tasks relevant to video analytics.

In edge surveillance systems, KD enables cameras equipped with ARM Cortex-A processors to run detection models that approach the accuracy of much larger cloud-deployed networks. Setyanto et al. [4] propose replacing YOLOv4’s CSPDarknet53 backbone with MobileNetV2 or RepViT, using KD from the original YOLOv4 teacher, and evaluating on Jetson Nano, Jetson Orin Nano, and Raspberry Pi 4B. Their results confirm that KD-compressed models enable accurate real-time detection on hardware that was previously unable to run any YOLO variant.

4.3 Adaptive Video Encoding and Frame Filtering

Not all video frames carry equal information. In a parking lot surveillance scenario, 98% of frames may be identical (empty lot); only frames with motion warrant expensive DNN inference. Adaptive frame filtering intelligently selects which frames to process and at what resolution, dramatically reducing average computational load without sacrificing detection quality on events of interest.

Reducto (Li et al., ACM SIGCOMM 2020) pioneered cheap feature differencing at the camera level to skip frames with low change scores. The Ke et al. survey (2023) [2] documents that systems like Reducto can achieve 3–6x reduction in frames sent to the edge server while retaining over 95% of the accuracy compared to full-frame processing.

For multi-stream deployments, bandwidth orchestration is equally critical. The BiSwift system (2023) [6] introduces a bilevel reinforcement learning algorithm that jointly controls per-stream bandwidth allocation and DNN pipeline selection. By adaptively selecting between three processing pipelines (lightweight, medium, full) based on frame content, BiSwift reduces bandwidth consumption by 40% compared to fixed-rate baselines while improving accuracy by 15%. Its adaptive hybrid encoder further enables aggressive video compression with anchor frames used to compensate for accuracy loss from compression artifacts.

Context-aware offloading is another dimension. Chen et al. (IEEE TMC, 2024) [9] formalize the resolution and deviation-based offloading (RDO) problem: given heterogeneous edge-device compute and dynamic network bandwidth, when should a device offload a frame to the edge server versus process it locally with a lightweight tracker? Their online learning approach maximizes inference accuracy and processing rate simultaneously without requiring ground-truth labels from previous inferences.

4.4 Task Offloading and Resource Scheduling

When an edge device cannot complete inference within a deadline, partial or complete offloading to an edge server or cloud becomes necessary. Offloading decisions must balance computation latency, communication latency, inference accuracy, energy consumption, and data privacy simultaneously.

ModelIO (Chen et al., NOSSDAV 2022) [10] frames video analytics pipeline control as a deep reinforcement learning problem. An RL agent jointly decides video preprocessing parameters, DNN model selection (from a library of models at different accuracy-complexity trade-offs), local vs. remote inference allocation, and offloading

targets. Trained in a simulated edge-cloud environment, ModelIO learns to maximize analytics accuracy while satisfying latency constraints.

The Infer-EDGE framework (2025) [11] uses an Advantage Actor-Critic RL agent to dynamically tune inference parameters (batch size, model variant, precision) on “just-in-time” edge nodes. The framework handles the fundamental tension that offloading raw video violates privacy, but keeping all inference on-device may exceed power and latency budgets. Infer-EDGE achieves Pareto-optimal trade-offs across latency, accuracy, and energy consumption on a hardware testbed.

DNN model partitioning—splitting a neural network at an intermediate layer and distributing execution across device and server—is an orthogonal approach. Neurosurgeon (Kang et al., 2017) established the paradigm; more recent works like PARTNNer (ACM TECS, 2024) [11] provide platform-agnostic adaptive partitioning that accounts for time-varying network conditions.

4.5 Hardware Accelerators

Dedicated silicon accelerators dramatically improve energy efficiency for DNN inference. Four families of accelerators are relevant to EVA: GPUs (NVIDIA Jetson family), NPU (Apple Neural Engine, ARM Ethos), FPGAs (Xilinx Versal, Intel Stratix), and ASICs (Google Edge TPU, Hailo-8).

Research by Soc-e (IEEE DCIS, 2022) [4] describes an FPGA-based EVA accelerator using Xilinx’s Deep Processing Unit (DPU). The Video Codec Unit (VCU) handles H.264/H.265 decode in the FPGA’s Programmable Logic (PL) region, while the DPU executes CNN inference. This tight hardware-software co-design achieves real-time 1080p inference at frame rates above 30 fps on a single FPGA card with a fraction of the power consumed by a server-grade GPU.

5G MEC integration further changes the hardware calculus. By co-locating edge compute nodes with 5G base stations (gNBs), operators achieve sub-millisecond local network latency, enabling new applications where camera-to-inference round trips must complete within a single video frame interval (33 ms at 30 fps).

5. Core Application Domains

The Xu et al. IEEE COMST survey (2023) [1] and the IoT journal review (Ke et al., 2023) [2] together catalog the principal application domains of EVA. We organize these into six categories.

5.1 Intelligent Transportation

Smart traffic management is among the most mature EVA applications. Edge cameras at intersections run vehicle detection and classification in real time, providing input to adaptive signal control systems. Object counting, speed estimation, license plate recognition, and incident detection (accidents, wrong-way driving) are the core analytics tasks. Edge-local processing is essential because traffic signal timing must respond within 100–200 ms; any cloud round-trip would violate this constraint.

Edge cameras also enable vehicle re-identification across intersections, which requires multi-camera collaborative analytics. The Anveshak framework (Khochare et al.) [2] demonstrated a tracking system that expands its camera network when an object is lost by one camera, relying on collaborative edge inference to maintain continuous tracks.

5.2 Public Safety and Surveillance

Crowd anomaly detection, intrusion detection, violent behavior recognition, and abandoned object detection are canonical public safety tasks. Deep learning models trained on large-scale video datasets (UCF-Crime, ShanghaiTech) now achieve high accuracy on these tasks, but require significant compute. EVA systems use lightweight two-branch architectures (spatial + temporal/optical flow) at the edge, with heavier transformer-based models reserved for cloud verification.

Privacy-by-design constraints are particularly acute here. Barthelemy et al. [2] designed an edge-based computer vision system for transportation monitoring that "ensures privacy of citizens" by performing all face and license plate analysis on-device, transmitting only anonymized aggregate statistics (vehicle counts, pedestrian density) to the cloud.

5.3 Industrial Automation and Quality Control

Manufacturing defect detection, assembly verification, and equipment monitoring are high-value EVA applications. These tasks require very high frame rates (often 60–120

fps) and fine-grained detection (millimeter-scale defects). FPGA-based accelerators are particularly well-suited here, offering deterministic latency and programmable vision pipelines.

Industrial vibration monitoring, as cited in edge AI deployment literature [5], uses structured pruning to achieve 40% faster inference with only 2% accuracy loss, enabling on-device anomaly detection without cloud dependence. This is critical in factory environments where network connectivity cannot be guaranteed.

5.4 Smart Agriculture

Precision agriculture deploys edge cameras on drones and ground robots to monitor crop health, detect plant diseases, count livestock, and estimate yield. The "smart farming" application domain, documented by Ke et al. [2], applies video analytics to "improve the quantity and quality of crops" by monitoring animal status and adjusting planting methods based on automated visual analysis. Drone-mounted edge devices must process video in real time with limited battery power, making aggressive model compression essential.

5.5 Healthcare and Assisted Living

Patient fall detection, gait analysis, activity monitoring in elderly care facilities, and surgical workflow analysis are emerging EVA applications. These applications impose strict privacy requirements (video of patients cannot leave the facility) and demand high reliability (missed falls can have fatal consequences). Edge-only architectures with locally trained, personalized models are the dominant approach.

5.6 Augmented and Extended Reality

AR/VR headsets and smart glasses require sub-20 ms pose estimation and semantic segmentation to provide convincing occlusion-correct overlays. Edge servers co-located in the same room or building provide the additional compute headroom needed when on-device inference is insufficient. Research on edge-assisted AR (MPCP system, EdgeSys 2023) [34] frames the pipeline scheduling problem as minimizing the staleness of server-returned inference results, which directly impacts the accuracy of AR overlays.

6. Benchmark Frameworks and Datasets

Systematic evaluation of EVA systems requires standardized datasets and evaluation frameworks. The following are the most widely used in recent literature.

Dataset / Framework	Domain	Key Characteristics	Common EVA Tasks
UCF-Crime	Surveillance	1,900 long untrimmed videos, 13 anomaly classes	Anomaly detection, action recognition
COCO / COCO-video	Object Detection	330K images, 80 categories, bounding boxes, keypoints	Object detection, segmentation benchmarking
VisDrone	UAV Surveillance	263 videos, dense crowds, small objects	Multi-object tracking, person detection
D2-City	Dashcam / Traffic	High occlusion, large object size variation	Vehicle detection, driver assistance
Vehicle detection, driver assistance	Anomaly Detection	330,165 frames, 13 scenes, 130 abnormal events	Frame-level anomaly scoring
Microsoft Rocket	Live Video Analytics	Open-source EVA pipeline and evaluation harness	System-level latency, accuracy benchmarking

Table 2: Key Datasets and Evaluation Frameworks for Edge Video Analytics Research

Hardware benchmarking is equally important. The EVA research community uses standardized edge hardware testbeds including NVIDIA Jetson TX2, Jetson AGX Xavier, Jetson Orin, Google Coral Dev Board (Edge TPU), and Intel Neural Compute Stick 2. Metrics reported across papers include frames per second (fps), inference

latency (ms), energy per inference (mJ/frame), and model size (MB), all reported together to enable multi-objective comparison.

7. Open Challenges

7.1 Accuracy-Efficiency Trade-Off

The Pareto frontier between model accuracy and inference efficiency remains the central challenge. Compression techniques inevitably sacrifice some accuracy; the question is how much. For safety-critical applications (e.g., intruder detection in nuclear facilities, pedestrian detection in autonomous vehicles), accuracy cannot be compromised below certain thresholds. The community lacks principled methods for specifying and guaranteeing accuracy-efficiency trade-offs across diverse edge hardware and video content distributions.

7.2 Hardware Heterogeneity

Edge deployments comprise enormously heterogeneous hardware: ARM CPUs, mobile GPUs, FPGAs, dedicated NPUs, and quantum-dot neural accelerators. A model optimized for NVIDIA Jetson via TensorRT may not execute efficiently on a Google Edge TPU or a Hailo-8 without re-compilation and re-calibration. The PArtNNer system [11] addresses platform-agnostic DNN partitioning, but a general solution for cross-platform model portability at scale remains elusive.

7.3 Concept Drift and Continuous Learning

Video analytics models trained on historical data degrade when the visual distribution shifts: new lighting conditions, seasonal changes in vegetation, changes in crowd demographics, or new threat patterns not seen during training. Efficient online retraining on edge devices without interrupting inference is an active research problem. The efficient online DNN inference with continuous learning framework (Li et al., IWQoS 2024) [11] addresses this through concept drift detection and cloud-assisted selective retraining, but the problem is far from solved.

7.4 Privacy and Security

Even edge-local processing raises privacy concerns: edge servers may be physically accessible to unauthorized parties, and intermediate feature tensors (even without raw pixels) may enable attribute inference attacks. Differential privacy, homomorphic encryption, and federated learning are being explored, but all impose significant computational overhead that is hard to absorb on resource-constrained edge hardware. The cloud-edge-terminal collaborative survey (2025) [9] highlights that "secure multi-party computation, homomorphic encryption, and differential privacy provide additional safeguards despite computational overhead."

7.5 Multi-Camera Coordination Complexity

As camera counts scale into the thousands, the complexity of multi-camera coordination explodes. Object re-identification across non-overlapping camera views requires large embedding models that cannot run at the edge. Scheduling overlapping fields of view to avoid redundant computation (CrossRoI [1]) requires global knowledge of camera network topology that is expensive to maintain dynamically.

7.6 Standardization and Evaluation Gaps

There is no universally accepted benchmark that jointly evaluates inference accuracy, latency, energy consumption, bandwidth usage, and privacy leakage for EVA systems. The IoT journal review (2023) [2] proposes 17 evaluation criteria but acknowledges that most papers report only 3–5. Without standardization, fair comparison across published systems is difficult.

8. Future Directions

8.1 Federated Learning at the Edge

Federated learning (FL) enables multiple edge cameras to collaboratively train a shared model without sharing raw video. Each device trains locally on its own data stream and shares only gradient updates (or model deltas) with an aggregation server. This provides both privacy guarantees and the ability to learn from diverse visual

distributions distributed across a city. Recent work on vertical and horizontal federated learning for surveillance suggests that FL can match centralized training accuracy within 2–3% while keeping all raw data on-device. The integration of FL with continuous learning and concept drift adaptation is an emerging frontier.

8.2 Transformer Architectures for Edge Video Understanding

Vision Transformers (ViT) and video transformers (TimeSformer, VideoMAE) have surpassed CNN-based models on standard action recognition benchmarks. However, their quadratic self-attention complexity makes them challenging to deploy at the edge. Research on efficient transformer variants (Swin Transformer, MobileViT, EfficientViT), token pruning, and mixed CNN-transformer architectures is accelerating. Within 2–3 years, transformer-based edge inference at 30 fps on mid-range edge hardware is likely to become standard.

8.3 Neural Architecture Search for Edge Deployment

Neural Architecture Search (NAS) automates the discovery of optimal network architectures for given hardware constraints. Hardware-aware NAS directly incorporates latency and energy measurements from target edge devices into the search objective. OFA (Once-for-All), EfficientDet-NAS, and AutoML frameworks can produce device-specific model variants significantly outperforming hand-crafted architectures. Tighter integration of NAS with EVA system co-design is expected to yield another wave of efficiency improvements.

8.4 5G and 6G Network Slicing Integration

5G network slicing enables operators to guarantee end-to-end latency, bandwidth, and reliability for specific application traffic classes. An EVA slice can be configured with guaranteed sub-5 ms device-to-MEC latency, enabling video analytics applications with extremely tight real-time requirements. 6G, expected to be commercially deployed by 2030, promises terabit wireless throughput and native AI integration in the radio access network, potentially enabling inference co-located with the antenna.

8.5 Generative AI at the Edge

Generative models (diffusion models, LLMs for video captioning) are increasingly being applied to video analytics tasks: generating natural-language descriptions of events, synthesizing training data for rare anomaly categories, and performing zero-shot or few-shot anomaly detection without labeled examples. Running generative inference at the edge remains challenging given model sizes in the billions of parameters, but quantized diffusion models (4-bit) running on NPUs are already feasible for offline applications. Real-time generative video captioning at the edge is a 5-year research horizon.

9. Conclusion

Edge Video Analytics has emerged as one of the most active and impactful intersections of systems research and computer vision. The field addresses a fundamental practical need: the vast majority of the world's camera-generated video cannot be economically or practically transmitted to the cloud for analysis, yet the deep learning models required for accurate analysis were designed for cloud-scale hardware.

The last five years have produced a rich toolkit for bridging this gap. Hierarchical edge-cloud architectures provide a flexible deployment substrate. Model compression techniques (pruning, quantization) can reduce DNN footprints by 10–50x with minimal accuracy loss. Knowledge distillation allows compact edge models to approach the accuracy of their large teacher counterparts. Adaptive frame filtering and bandwidth orchestration reduce the volume of data requiring expensive inference. Reinforcement learning-based offloading schedulers dynamically navigate latency-accuracy-energy trade-offs in real time.

Despite this progress, significant challenges remain. The accuracy-efficiency frontier must be pushed further to support safety-critical EVA applications. Hardware heterogeneity demands better model portability tools. Concept drift requires practical continuous learning solutions. Privacy and security at the edge need cryptographic guarantees with acceptable overhead. Standardized, multi-dimensional benchmarking frameworks are needed to enable fair comparison.

As 5G/6G networks mature, edge hardware becomes more powerful, and AI frameworks become better optimized for heterogeneous silicon, the economic and technical case for EVA will continue to strengthen. The research community is well-positioned to meet these challenges, and the application impact—safer cities, smarter industries, more accessible healthcare—justifies sustained investment.

References

All references are to peer-reviewed publications within the 2020–2025 window unless otherwise noted.

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